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CYCLICITY AND EXTRACTION DOMAINS

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Abstract. This paper attempts to provide a minimalist analysis of CED effects (see Huang 1982) in terms of derivational dynamics in a cyclic system. Assuming Uriagereka's (1999) Multiple Spell-out system, we argue that CED effects arise when a syntactic object K that is required at a given derivational step has become inaccessible to the computational system at a previous derivational stage, when the chunk of structure containing K was spelled out. Assuming Nunes's (1995, 1998) analysis of parasitic gaps in terms of sideward movement, we argue that standard parasitic gap constructions do not exhibit CED effects because K manages to move to a different derivational workspace before the structure containing it is spelled out. Finally, we provide an account of the cases where parasitic gap constructions appear to show CED effects by relying on cyclic access to the numeration, along the lines proposed by Chomsky (1998).

1. Introduction

If something distinguishes the Minimalist Program of Chomsky (1995, 1998) from other models within the principles-and-parameters framework, that is the assumption that the language faculty is an optimal solution to legibility conditions imposed by external systems. From this perspective, a main desideratum of the program is to derive substantive principles from interface ("bare output") conditions, and formal principles from economy conditions. It is thus natural that part of the minimalist agenda is devoted to reevaluating the theoretical apparatus developed within the principles-and-parameters framework, with the goal of explaining on more solid conceptual grounds the wealth of empirical material uncovered in the past decades. This paper takes some steps towards this goal by deriving Condition on Extraction Domains (CED) effects (in the sense of Huang 1982) in consonance with these general minimalist guidelines.

Within the principles-and-parameters framework, the CED is generally assumed to be a government-based locality condition that restricts movement operations (see Huang 1982 and Chomsky 1986, for instance). But once the notion of government is abandoned in the Minimalist Program, as it involves nonlocal relations (see Chomsky 1995:chap. 3), the data that were accounted for in terms of the CED call for a more principled analysis.

Some of the relevant data regarding the CED are illustrated in examples (1)–(3). Example (1) shows that regular extraction out of a subject or an adjunct yields unacceptable results; (2) shows that parasitic gap constructions

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structurally analogous to (1) are much more acceptable; finally, (3) shows that if the licit parasitic gaps of (2) are further embedded within a CED island such as an adjunct clause, unacceptable results arise again (see Kayne 1984, Contreras 1984, Chomsky 1986).

(1) a. *[$_{CP}$ [which politician]$_i$ [$_{C'}$ did+Q [$_{IP}$ [pictures of t_i] upset the voters]]]
b. *[$_{CP}$ [which paper]$_i$ [$_{C'}$ did+Q [$_{IP}$ you read *Don Quixote* [$_{PP}$ before filing t_i]]]]
(2) a. [$_{CP}$ [which politician]$_i$ [$_{C'}$ did+Q [$_{IP}$ [pictures of pg_i] upset t_i]]]
b. [$_{CP}$ [which paper]$_i$ [$_{C'}$ did+Q [$_{IP}$ you read t_i [$_{PP}$ before filing pg_i]]]]
(3) a. *[$_{CP}$ [which politician]$_i$ [$_{C'}$ did+Q [$_{IP}$ you criticize t_i [$_{PP}$ before [pictures of pg_i] upset the voters]]]]
b. *[$_{CP}$ [which book]$_i$ [$_{C'}$ did+Q [$_{IP}$ you finally read t_i [$_{PP}$ after leaving the bookstore [$_{PP}$ without finding pg_i]]]]]

Thus far, the major locality condition explored in the Minimalist Program is the Minimal Link Condition stated in (4) (see Chomsky 1995:311).

(4) *Minimal Link Condition*
K attracts α only if there is no β, β closer to K than α, such that K attracts β.

The unacceptability of (5a), for instance, is taken to follow from a Minimal Link Condition violation: at the derivational step represented in (5b), the interrogative complementizer Q should have attracted the closest *wh*-element *who*, instead of attracting the more distant *what*.

(5) a. *[I wonder [$_{CP}$ what$_i$ [$_{C'}$ Q [$_{IP}$ who [$_{VP}$ bought t_i]]]]]
b. [$_{CP}$ Q [$_{IP}$ who [$_{VP}$ bought what]]]

The Minimal Link Condition is in consonance with the general economy considerations underlying minimalism, in that it reduces the search space for computations, thereby reducing ("operative") computational complexity. However, it has nothing to say about CED effects such as the ones illustrated in (1)–(3). In (1a), for instance, there is no *wh*-element other than *which politician* that Q could have attracted.

In this paper we argue, first, that CED effects arise when a syntactic object that is required at a given derivational step has become inaccessible to the computational system at a previous derivational stage; and second, that the contrasts between (1) and (2), on the one hand, and between (2) and (3), on the other, are due to their different derivational histories. These results arise as by-products of two independent lines of research on the role of Kayne's (1994) Linear Correspondence Axiom (LCA) in the minimalist framework: Uriagereka's (1999) Multiple Spell-out system, which derives the induction

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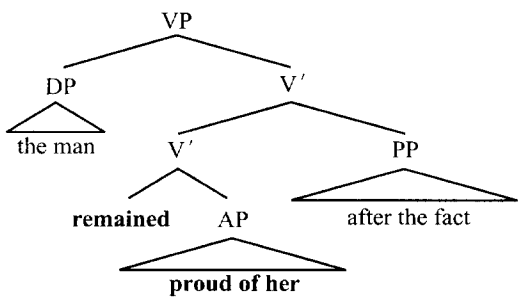
step of the LCA by eliminating the unmotivated stipulation that Spell-out must apply only once; and Nunes's (1995, 1998) version of the copy theory of movement, which permits instances of sideward movement (i.e., movement between two unconnected syntactic objects) if the LCA is satisfied.

The paper is organized as follows. In section 2, we show how the standard CED effects illustrated in (1) can be accounted for within Uriagereka's (1999) Multiple Spell-out theory. In section 3, we show that sideward movement allows constrained instances of movement from CED islands, resulting in parasitic gap constructions such as (2). In section 4, we provide an account of the unacceptability of constructions such as (3) by reducing the computational complexity associated with sideward movement in terms of Chomsky's (1998) cyclic access to subarrays. Finally, a brief conclusion is presented in section 5.

2. Basic CED Effects

Any account of the CED has to make a principled distinction between complements and noncomplements (see Cattell 1976 for early, very useful discussion). Kayne's (1994) LCA has the desired effect: a given head can be directly linearized with respect to the lexical items within its complement, but not with respect to the lexical items within its subject or adjunct. The reason is trivial. Consider the phrase marker in (6), for instance (irrelevant details omitted).

(6)



It is a simple fact about the Merge operation that only the terminal elements in boldface in (6) can be assembled without ever abandoning a single derivational workspace; by contrast, the terminal elements under DP and PP must first be assembled in a separate derivational space before being connected to the rest.

One can capitalize on this derivational fact in various ways. Let us recast Kayne's (1994) LCA in terms of Chomsky's (1995) bare phrase structure

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and simplify its definition by eliminating the recursive step, as formulated in (7).[1]

(7) *Linear Correspondence Axiom*
A lexical item α precedes a lexical item β iff α asymmetrically c-commands β.

Clearly, all the terminals in boldface in (6) stand in valid precedence relations, according to (7). The question is how they can establish precedence relations with the terminals within DP and PP, if the LCA is as simple as (7).

Uriagereka (1999) suggests an answer, by taking the number of applications of the rule of Spell-out to be determined by standard economy considerations and not by the unmotivated stipulation that Spell-out must apply only once. Here we will focus our attention to cases where multiple applications of Spell-out are triggered by linearization considerations (see Uriagereka 1999 for other cases and further discussion). The reasoning goes as follows. Let us refer to the operation that maps a phrase structure into a linear order of terminals in accordance with the LCA in (7) as *Linearize*.[2] Under the standard assumption that phrasal syntactic objects are not legitimate objects at the PF level, Linearize can be viewed as an operation imposed on the phonological component by legibility requirements of the articulatory-perceptual interface, as essentially argued by Higginbotham (1983). If this is so and if the LCA is as simple as (7), the computational system should not ship complex structures such as (6) to the phonological component by means of the Spell-out operation, because Linearize would not be able to determine precedence relations among all the lexical items. Assuming that failure to yield a total ordering among lexical items leads to an illicit derivation, the system is forced to employ multiple applications of Spell-out, targeting chunks of structure that Linearize can operate with.

Under this view, the elements in subject and adjunct position in (6) can be linearized with respect to the rest of the structure in accordance with (7) in the following way: (i) the DP and the PP are spelled out separately and in the phonological component, their lexical items are linearized internal to them; and (ii) the DP and the PP are later "plugged in" where they belong in the whole structure. We assume that the label of a given structure provides the "address" for the appropriate plugging in, in both the phonological and the interpretive components.[3] That is, applied to the syntactic object K = {γ, {α,

[1] For purposes of presentation, we ignore cases where two heads are in mutual c-command. For discussion, see Chomsky 1995 (p. 337).

[2] In Chomsky 1995 (chap. 4), the term *LCA* is used to refer both to the Linear Correspondence Axiom and the mapping operation that makes representations satisfy this axiom, as becomes clear when it is suggested that the LCA may delete traces (see Chomsky 1995:337). We will avoid this ambiguity and use the term *Linearize* for the operation.

[3] See Uriagereka (1999) for a discussion of how agreement relations could also be used as addresses for spelled-out structures.

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β}}, with label γ and constituents α and β (see Chomsky 1995:chap. 4). Spell-out ships {α, β} to the phonological and interpretative components, leaving K only with its label. Because the label encodes the relevant pieces of information that allow a category to undergo syntactic operations, K itself is still accessible to the computational system, despite the fact that its constituent parts are, in a sense, gone: thus, for instance, K can move and is visible to linearization when the whole structure is spelled-out. Another way to put it is to say that once the constituent parts of K are gone, the computational system treats it as a lexical item. In order to facilitate keeping track of the computations in the following discussion, we use the notation K = $[_{\gamma}$ <α, β>] to represent K after it has been spelled out.

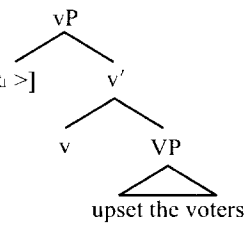
An interesting consequence of this proposal is that multiple Spell-out of separate derivational cascades derives Cattell's (1976) original observation that only complements are transparent to movement. When Spell-out applies to the subject DP in (6), for instance, the computational system no longer has access to its constituents and, therefore, no element can be extracted out of it. Let us consider a concrete case by examining the relevant details of the derivation of (8), after the stage where the structures K and L in (9) have been assembled by successive applications of Merge.

(8) *Which politician did pictures of upset the voters?
(9) a. K = $[_{vP}$ upset the voters]
b. L = [pictures of which politician]

If the LCA is as simple as in (7), the complex syntactic object resulting from the merger of K and L in (9) would not be linearizable because the constituents of K would not enter into a c-command relation with the constituents of L. The computational system then applies Spell-out to L, allowing its constituents to be linearized in the phonological component, and merges the spelled-out structure L′ with K, as illustrated in (10).[4]

(10)

L′ = $[_{pictures}$ < pictures, of, which, politician >]



[4] Following Uriagereka (1999), we assume that spelled-out structures do not project. Hence, if the computational system applies Spell-out to K instead of L in (9), the subsequent merger of L and the spelled-out K does not yield a configuration for the appropriate thematic relation to be established, violating the θ-Criterion. Similar considerations apply, *mutatis mutandis*, to spelling out the target of adjunction instead of the adjunct in example (14)

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Further computations involve the merger of *did* and movement of L′ to [Spec,TP]. Assuming Chomsky's (1995:chap. 3) copy theory of movement, this amounts to saying that the computational system copies L′ and merges it with the assembled structure, yielding the structure in (11) (the deletion of the lower copy in the phonological component is discussed in section 3).

(11) $[_{TP}$ $[_{pictures}$ <pictures, of, which, politician>] $[_{T'}$ did $[_{vP}$ $[_{pictures}$ <pictures, of, which, politician>] $[_{v'}$ upset the voters]]]]

In the next steps, the interrogative complementizer Q merges with TP and *did* adjoins to it, yielding (12).

(12) $[_{CP}$ did+Q $[_{TP}$ $[_{pictures}$ <pictures, of, which, politician>] $[_{T'}$ did $[_{vP}$ $[_{pictures}$ <pictures, of, which, politician>] $[_{v'}$ upset the voters]]]]]

In (12), there is no element that can check the strong *wh*-feature of Q. Crucially, the *wh*-element of either copy of L = $[_{pictures}$ <pictures, of, which, politician>] became unavailable to the computational system after L was spelled out. The derivation therefore crashes. Under this view, there is no way for the computational system to yield the sentence in (8) if derivations unfold in a strictly cyclic fashion, as we are assuming. To put it in more general terms, extraction out of a subject is prohibited because, at the relevant derivational point, there is literally no syntactic object within the subject that could be copied.

Similar considerations apply to the sentence in (13), which illustrates the impossibility of "extraction" out of an adjunct clause.

(13) *Which paper did you read *Don Quixote* before filing?

Assume, for concreteness, that the temporal adjunct clause of (13) is adjoined to vP. Once K and L in (14) have been assembled, Spell-out must apply to L, before K and L merge; otherwise, the lexical items of K could not be linearized with respect to the lexical items of L. After L is spelled out as L′, it merges with K, yielding (15). In the phonological component, Linearize applies to the lexical items of L′ and the resulting sequence will be later plugged into the appropriate place, after the whole structure is spelled out. The linear order between the lexical items of L and the lexical items of K will then be (indirectly) determined by whatever fixes the order of adjuncts in the grammar.[5]

[5] That is, regardless of whether adjuncts are linearized by the procedure that linearizes specifiers and complements or by a different procedure (see Kayne 1994 and Chomsky 1995 for different views), the important point to keep in mind is that, if the formulation of the LCA is to be as simple as (7), the lexical items within L′ in (15) cannot be directly linearized with respect to the lexical items contained in the lower vP segment

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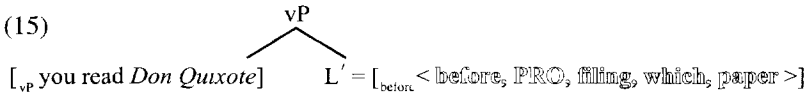
(14) a. K = [$_{vP}$ you read *Don Quixote*]
b. L = [$_{PP}$ before PRO filing which paper]

(15)



What is relevant for the current discussion is that after the (simplified) structure in (16) is formed, there is no *wh*-element available to check the strong *wh*-feature of Q and the derivation crashes; in particular, *which paper* is no longer accessible to the computational system at the step where it should be copied to check the strong feature of Q. As before, the sentence in (13) is underivable through the cyclic derivation outlined in (14)–(16).

(16) [$_{CP}$ did+Q [$_{TP}$ you [$_{vp}$ [$_{vP}$ read *Don Quixote*] [$_{before}$ <before, PRO, filing, which, paper>]]]]

Finally, let us consider (17a). Structures like (17a) have recently been taken to show that cyclicity cannot be violated. If movement of *who* to [Spec,CP] were allowed to proceed prior to the movement of α to the subject position, (17a) should pattern like (17b), where *who* is extracted from within the object, contrary to fact. If cyclicity is inviolable, so the argument goes, *who* in (17a) must have moved from within the subject, yielding a CED effect (see Chomsky 1995:328, Kitahara 1997:33).

(17) a *who$_i$ was [$_\alpha$ a picture of t_i]$_k$ taken t_k by Bill
b. who$_i$ did Bill take [$_\alpha$ a picture of t_i]

A closer examination of this reasoning however reveals that it only goes through in a system that takes traces to be grammatical primitives. If the trace of α in (17a) is simply a copy of α, as shown in (18), the copy of *who* inside the object should in principle be able to move to [Spec,CP], incorrectly yielding an acceptable result. Crucially, the copy of *who* within the subject does not c-command the copy within the object and no intervention effect should arise.

(18) [$_{CP}$ Q [$_{TP}$ [$_\alpha$ a picture of who] was taken [$_\alpha$ a picture of who] by Bill]]

Before we discuss how the system we have been exploring, which assumes the copy theory of movement, is able to account for the unacceptability of (17a), let us first consider the derivation of (19), where no *wh*-movement is involved.

(19) Some pictures of John were taken by Bill.

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In (20), the computational system makes a copy of *some pictures of John*, spells it out, and merges the spelled-out copy with K, forming the object in (21).

(20) a. K = [$_{TP}$ were [$_{VP}$ taken [some pictures of John] by Bill]]
b. L = [$_{some}$ <some, pictures, of, John>]

(21) [$_{TP}$ [$_{some}$ <some, pictures, of, John>] [$_{T'}$ were [$_{VP}$ taken [some pictures of John] by Bill]]]

Under reasonable assumptions regarding chain uniformity, the elements in subject and object positions in (21) cannot constitute a chain because they are simply different kinds of syntactic objects (a label and a phrasal syntactic object). Assume for the moment that lack of chain formation in (21) leads to a derivational crash (see the next section for further discussion). Given the perfect acceptability of (19), an alternative route must be available.

Recall that under the Multiple Spell-out approach, the number of applications of Spell-out is determined by economy. Thus, complements in general do not need to be spelled out in separate derivational cascades because they can be linearized within the derivational cascade involving the subcategorizing verb—that is, a single application of Spell-out can linearize both the verb and its complement. In the case of (21), however, a licit chain can only arise if the NP in the object position has been independently spelled out, so that the two copies can constitute a chain. This leads us to conclude that convergence demands may force Spell-out to apply to complements as well.

That being so, the question then is whether the object is spelled out in (20a) before copying takes place or only after the structure in (21) has been assembled. Again, we may find the answer in economy: if Spell-out applies to *some pictures of John* before it is copied, the copies will be already spelled out and no applications of Spell-out will be further required for the copies. The derivation of (19) therefore proceeds along the lines of (22): the NP is spelled out before being copied in (22a) and its copy merges with the whole structure, as shown in (22b); the two copies of the NP can then form a licit chain and the derivation converges.

(22) a. [$_{TP}$ were [$_{VP}$ taken [$_{some}$ <some, pictures, of, John>] by Bill]]
b. [$_{TP}$ [$_{some}$ <some, pictures, of, John>] [$_{T'}$ were [$_{VP}$ taken [$_{some}$ <some, pictures, of, John>] by Bill]]]

Returning to (17a), its derivation proceeds in a cyclic fashion along the same lines, yielding the (simplified) structure in (23). Once the stage in (23) is reached, no possible continuation results in a convergent derivation: the strong *wh*-feature of Q must be checked and neither copy of *who* is accessible to the computational system. The approach we have been exploring here is therefore able to account for the unacceptability of (17a), while still adhering to the view that traces are simply copies and not grammatical formatives.

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(23) [$_{CP}$ was+Q [$_{TP}$ [$_{a}$ <a, picture, of, who>] [$_{VP}$ taken [$_{a}$ <a, picture, of, who>] by Bill]]]

To summarize, CED effects arise when a given syntactic object K that would be needed for computations at a derivational stage D_n has been spelled out at a derivational stage D_1 prior to D_n, thereby becoming inaccessible to the computational system after D_1. Under this view, the CED is not a primitive condition on movement operations; rather, it presents itself as a natural consequence in a derivational system that obeys strict cyclicity and takes general economy considerations to determine the number of applications of Spell-out.[6]

The question that we now face is how to explain the complex behavior of parasitic gap constructions with respect to the CED, as seen in the introduction, if the explanation for the CED developed above is correct. This is the topic of the next sections. Notice, for instance, that we cannot simply assume that parasitic gap constructions bypass some condition X that regular extractions obey; in fact, we are suggesting that there is no particular condition X to prevent extraction and, therefore, no way to bypass it either. Before going into the analysis proper, we briefly review Nunes's (1995, 1998) analysis of parasitic gaps in terms of sideward movement, which provides us with the relevant ingredients to address the issue of CED effects in parasitic gap constructions.

3. Sideward Movement and CED Effects

With the incorporation of the copy theory into the Minimalist Program, Move has been conceived of as a complex operation encompassing: (i) a suboperation of copying; (ii) a suboperation of merger; (iii) a procedure identifying copies as chains; and (iv) a suboperation deleting traces (lower copies) for PF purposes (see Chomsky 1995:250). Nunes (1995, 1998) develops an alternative version of the copy theory of movement with two main distinctive features.

First, his theory takes deletion of traces in the phonological component to be prompted by linearization considerations. Take the structure in (24b), for instance, which is based on the (simplified) initial numeration N in (24a) and arises after *John* moves to the subject position.

(24) a. N = {*arrested*$_1$, *John*$_1$, *was*$_1$}
b. [John^1 [was [arrested John^1]]]

[6] The approach outlined above is incompatible with a Larsonian analysis of double-object constructions (see Larson 1988), if extraction from within a direct object in a ditransitive construction is to be allowed.

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The two occurrences of *John* in (24b) are *nondistinct copies* (henceforth represented by superscripted indices) in the sense that both of them arise from the same item within N in (24a). If nondistinct copies are truly "the same" for purposes of linearization, (24b) cannot be mapped into a linear order [7] Given that the verb *was*, for instance, asymmetrically c-commands the lower copy of *John* and is asymmetrically c-commanded by the higher copy, the LCA should require that *was* precede and be preceded by *John*, which violates the asymmetry condition on linear orders (if α precedes β, it must be the case that β does not precede α). The attempted linearization of (24b) also violates the irreflexivity condition on linear orders (if α precedes β, it must be the case that $\alpha \neq \beta$): because the upper copy of *John* asymmetrically c-commands the lower one, *John* would be required to precede itself. Simply put, deletion of traces in the phonological component is forced upon a given chain CH in order for the structure containing CH to be linearized.[8]

The second distinctive feature of Nunes's (1995, 1998) version of the copy theory, which is crucial for the following discussion, is that Move is not taken to be a primitive operation of the computational system: it is rather analyzed as the mere reflex of the interaction among the independent operations described in (i)–(iv) above. In particular, this system allows constrained instances of *sideward movement*, where the computational system copies a given constituent α of a syntactic object K and merges α with a syntactic object L, which has been independently assembled and is unconnected to K, as illustrated in (25).[9]

(25) a. $[_K \ldots \alpha^1 \ldots]$ $\quad \alpha^1$ $\quad$ Merge $\quad [_L \ldots]$

Copy

b. $[_K \ldots \alpha^1 \ldots]$ $\quad [_M\ \alpha^1\ [_L \ldots]\ ]$

Let us consider how a parasitic gap construction such as (26a) can be derived under a sideward movement analysis, assuming that its initial numeration is the one given in (26b) (irrelevant items are omitted).

[7] The computation of nondistinct copies as the same for purposes of linearization may be taken to follow from Uriagereka's (1998) Conservation Law, according to which items in the numeration input must be preserved in the interpretive outputs.

[8] Notice that the structure in (24b) could also be linearized if the head of chain were deleted Nunes (1995, 1999) argues that the choice of the links to be deleted is actually determined by optimality considerations. Roughly speaking, the head of a chain in general becomes the optimal link with respect to phonetic realization as it participates in more checking relations. For the sake of presentation, we will assume that deletion always targets traces.

[9] The sequence of derivational steps in (25) has also been called *inter-arboreal operation* by Bobaljik & Brown (1997) and *paracyclic movement* by Uriagereka (1998).

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(26) a. Which paper did John file after reading?
b. N = {$which_1$, $paper_1$, did_1, $John_1$, PRO_1, Q_1, $file_1$, $after_1$, $reading_1$, v_2, C_1}

Example (27) shows the step after the numeration N in (26b) has been reduced to N′ and K has been assembled. Following Munn (1994) and Hornstein (1998), we assume that what Chomsky (1986) took to be null operator movement in parasitic gap constructions is actually movement of a syntactic object built from the lexical items of the numeration. From the perspective we are exploring, that amounts to saying that the computational system spells out *which paper* in (27b), makes a copy of the spelled-out object, and merges it with K to check whatever feature is involved in successive cyclic A′-movement, yielding L in (28a). The computational system then selects the preposition *after* and merges with L, forming the PP in (28b).

(27) a. N′ = {$which_0$, $paper_0$, did_1, $John_1$, PRO_0, Q_1, $file_1$, $after_1$, $reading_0$, v_1, C_0}
b. K = [$_{CP}$ C PRO reading [which paper]]

(28) a. L = [$_{CP}$ [$_{which}$ <which, paper>]^1 C PRO reading [$_{which}$ <which, paper>]^1]
b. M = [$_{PP}$ after [$_{CP}$ [$_{which}$ <which, paper>]^1 C PRO reading [$_{which}$ <which, paper>]^1]]

Consider now the stage after *file* is selected from the numeration, as shown in (29). Following Chomsky (1998), we assume that the selectional/thematic properties of *file* must be checked under Merge. However, possible continuations of the derivational step in (29) that merge *file* with the remaining elements of the reduced numeration N′ in (27a) do not lead to a convergent derivation; under standard assumptions, *John* should not be able to enter into a θ-relation with both *file* and the remaining light verb, or check both the accusative Case associated with the light verb and the nominative Case associated with *did*. Once lexical insertion leads to crashing, the system must resort to (sideward) movement, copying *which paper* from L and merging it with *file*, as shown in (30).[10] The *wh*-copy in (30b) may then "mind its own business" within derivational workspace P, independently of the other copies inside M. This is the essence of the account of parasitic gaps in terms of sideward movement.

(29) a. M = [$_{PP}$ after [$_{CP}$ [$_{which}$ <which, paper>]^1 C PRO reading [$_{which}$ <which, paper>]^1]]
b. O = file

[10] Recall that the label of a spelled-out object encodes the information that is relevant to the computational system; this includes the information that is required for a thematic relation to be established between *file* and [$_{which}$] in (30b).

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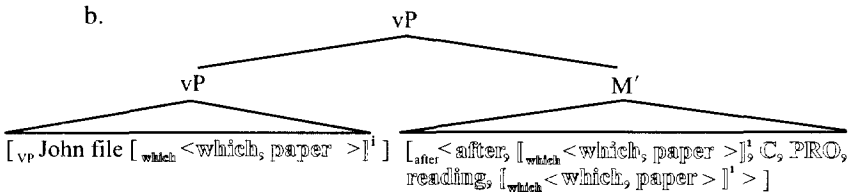
(30) a. M = [$_{PP}$ after [$_{CP}$ [$_{which}$ <which, paper>]^{1} C PRO reading [$_{which}$ <which, paper>]^{1}]]
b. P = [$_{VP}$ file [$_{which}$ <which, paper>]^{1}]

It is important to note that sideward movement of [$_{which}$ <which, paper>] in (29)–(30) was possible because M had not been spelled out; hence, the computational system had access not only to M itself but also to *the constituents of M*. The situation changes in subsequent derivational steps. As discussed in section 2, a complex adjunct must be spelled out before it merges with a given syntactic object; hence, the computational system spells out M as M′ in (31a) and merges M′ with the matrix vP, as represented in (31b).

(31) a. M′ = [$_{after}$ <after, [$_{which}$ <which, paper>]^{i}, C, PRO, reading, [$_{which}$ <which, paper>]^{i} >]
b.



Further computations involve lexical insertion of the remaining items of the numeration and movement of *John* and *did*, resulting in the (simplified) structure represented in (32).

(32) [$_{CP}$ did+Q [$_{IP}$ John [$_{vP}$ [$_{vP}$ file [$_{which}$ <which, paper>]^{1}] [$_{after}$ <after, [$_{which}$ <which, paper>]^{i}, C, PRO, reading, [$_{which}$ <which, paper>]^{i} >]]]]

The copies of [$_{which}$ <which, paper>] inside the adjunct clause in (32) are not available for copying, because the whole adjunct clause has already been spelled out. However, the copy in the object of *file* is still available to the computational system and, therefore, it can move to check the strong *wh*-feature of Q, yielding the (simplified) structure in (33), where the copies are numbered for ease of reference.

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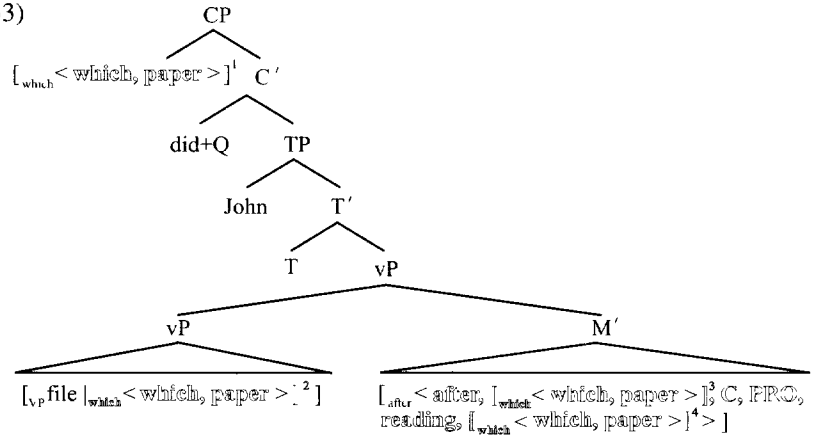
(33)



Let us now focus on the computations related to the deletion of *wh*-traces of (33) in the phonological component. As discussed before, the presence of multiple nondistinct copies prevents linearization. In the phonological component, the trace of the *wh*-chain within M is then deleted before Linearize applies to M to yield M′, as shown in (34).

(34) M′ = [$_{after}$ <after, [$_{which}$ <which, paper>]^3, C, PRO, reading, [~~which <which, paper>~~]^4 >]

After Spell-out applies to the whole structure in (33) and the previously spelled-out material is appropriately plugged in, two *wh*-chains should be further identified for trace deletion to take place: the "regular" chain CH_1 = ($copy_1$, $copy_2$), and the "parasitic" chain CH_2 = ($copy_1$, $copy_3$).[11] Identification of CH_1 is trivial because $copy_1$ clearly c-commands $copy_2$; hence, deletion of $copy_2$ is without problems. Identification of CH_2 is less obvious, because M is no longer a phrase structure after being linearized. However, if c-command is obtained by the composition of the elementary relations of sisterhood and containment, as proposed by Chomsky (1998:31) (see also Epstein 1995), $copy_1$ does c-command $copy_3$ in (33), because the sister of $copy_1$, namely C′, ends up containing $copy_3$ after the linearized material of M is properly plugged in.[12] The phonological component then deletes $copy_3$, yielding (35). Finally, Linearize applies to (35) and the PF output associated with (26a) is derived.[13]

[11] See Brody 1995 for a discussion of this kind of "forking" chain from a representational point of view.

[12] See the technical discussion about the structure of linearized objects by Uriagereka (1999), who shows that constituents of linearized objects such as $copy_3$ in (33) come out as terms in the sense of Chomsky (1995:chap. 4).

[13] As for the computation of the *wh*-copies inside the adjunct in (33) with respect to the whole structure in the interpretative component, there are two plausible scenarios to consider. In the first

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(35) [$_{CP}$ [$_{which}$ <which, paper>]^1 did+Q [$_{IP}$ John [$_{vP}$ [$_{vP}$ file ~~[$_{which}$ <which, paper>]^2~~] [$_{after}$ after, <~~[$_{which}$ <which, paper>]^3~~, C, PRO, reading, ~~[$_{which}$ <which, paper>]^4~~ >]]]]

Assuming that derivations proceed in such a strictly cyclic fashion, the contrast between unacceptable constructions involving "extraction" from within an adjunct island such as (13) and parasitic gap constructions such as (26a), therefore, follows from their different derivational histories. In the unacceptable case, the clausal adjunct has already been spelled out and its constituents are no longer available for copying at the derivational step where Last Resort would license the required copying (see section 2). In the acceptable parasitic gap constructions, on the other hand, a legitimate instance of copying takes place before the clausal adjunct is spelled out (see (29)–(30)); that is, sideward movement, if appropriately constrained by Last Resort, provides a kind of escape hatch for movement from within adjuncts.[14]

Similar considerations apply to parasitic gaps inside subjects. Let us consider the derivation of (36a), for instance, which starts with the numeration N in (36b).

(36) a. Which politician did pictures of upset?
b. N = {*which*$_1$, *politician*$_1$, *did*$_1$, *pictures*$_1$, *of*$_1$, *upset*$_1$, Q_1, v_1}

Suppose that after the derivational step in (37) is reached, K and L merge. No convergent result would then arise because there would be no element in the numeration N′ in (37a) to receive the external θ-role assigned by the light verb to be later introduced; additionally, if either K or the *wh*-phrase within K moved to [Spec,vP], they would be involved in more than one θ-relation within the same derivational workspace, leading to a violation of the θ-Criterion.[15]

(37) a. N′ = {*which*$_0$, *politician*$_0$, *did*$_1$, *pictures*$_0$, *of*$_0$, *upset*$_0$, Q_1, v_1}
b. K = [pictures of [which politician]]
c. L = upset

one, the interpretative component holds the spelled-out structures in a buffer and only computes chain relations after the whole structure is spelled out and the previously spelled-out structures are plugged in where they belong, in this case, identification of chains in terms of c-command is straightforward, because the structural relations have not changed. In the second scenario, the interpretative component operates with each object it receives, one at a time, and chain relations must then be determined in a paratactic-like fashion through the notion of antecedence The reader is referred to Uriagereka (1999) for general discussion of these possibilities.

[14] See Hornstein 1998 for a similar analysis.

[15] This is arguably what excludes the parasitic gap construction in (i), because sideward movement of *who* places it in two thematic configurations within the same derivational workspace.

(i) *who$_i$ did you give pictures of e_i to e_i

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The computational system may instead spell out the *wh*-phrase, make a copy of the spelled-out object, and merge it with *upset* (an instance of sideward movement), as shown in (38). Each copy of *which politician* in (38) will now participate in a θ-relation but in a different derivational workspace, as in (30).

(38) a. K = [pictures of $[_{which}$ <which, politician> $]^1$]
b. M = [upset $[_{which}$ <which, politician> $]^1$]

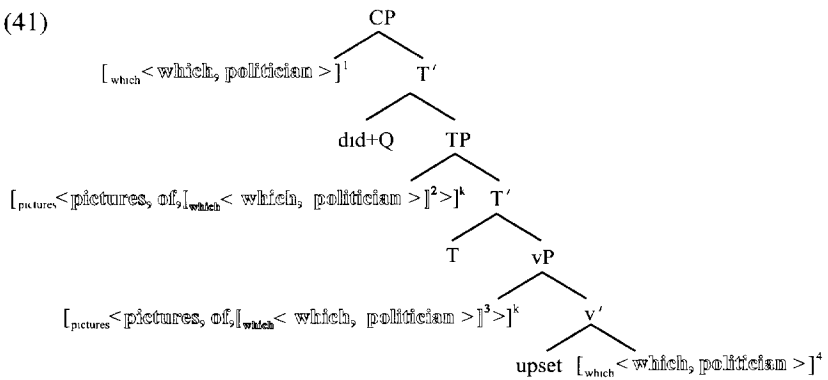
In the next steps, the light verb is selected from the numeration N′ in (37a) and merges with M in (38b), and the resulting structure merges with K after K is spelled out, yielding the (simplified) structure in (39). Further computations then involve merger and movement of *did*, and movement of the spelled-out subject to [Spec,TP], forming the (simplified) structure in (40).

(39) $[_{vP}$ $[_{pictures}$ <pictures, of, $[_{which}$ <which, politician> $]^i$ >] $[_{v'}$ upset $[_{which}$ <which, politician> $]^1$]]

(40) $[_{CP}$ did+Q $[_{TP}$ $[_{pictures}$ <pictures, of, $[_{which}$ <which, politician> $]^i$ > $]^k$ T $[_{vP}$ $[_{pictures}$ <pictures, of, $[_{which}$ <which, politician> $]^i$ > $]^k$ $[_{v'}$ upset $[_{which}$ <which, politician> $]^1$]]]

Among the three copies of *which politician* represented in (40), only the one in the object position of *upset* is available for copying; the other two became inaccessible after K in (37) was spelled out. The computational system then makes a copy of the accessible *wh*-element and merges it with the structure in (40), allowing Q to have its strong feature checked and finally yielding the structure in (41).

(41)



In the phonological component, deletion of the trace of the chain involving [Spec,TP] and [Spec,vP] in (41) ends up deleting $copy_3$ because $copy_3$ sits within [Spec,vP]. As for the other *wh*-copies, because $copy_1$ c-commands both $copy_2$ and $copy_4$ after the linearized material is plugged in (see

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discussion above), the chains $CH_1 = (copy_1, copy_2)$ and $CH_2 = (copy_1, copy_4)$ can be identified and their traces are deleted, yielding (42). (42) is then linearized and surfaces as (36a). Again, an apparent extraction from within a subject was only possible because Last Resort licensed sideward movement before the computational system spelled out the would-be subject.

(42) $[_{CP}\ [_{which}$ <which, politician> $]^1$ did+Q $[_{TP}\ [_{pictures}$ <pictures, of, ~~$[_{which}$ <which, politician> $]^2$~~ > $]^k$ T $[_{vP}\ [_{pictures}$ ~~<pictures, of, $[_{which}$ <which, politician> $]^3$ > $]^k$~~ $[_{v'}$ upset $[_{which}$ ~~<which, politician> $]^4$~~]]]

Although sideward movement may permit circumvention of CED islands in the cases discussed above, its output is constrained by linearization, like any standard instance of upward movement. That is, the same linearization considerations that trigger deletion of traces are responsible for ruling out unwanted instances of sideward movement (see Nunes 1995, 1998 for discussion). Take the derivation sketched in (43)–(45), for instance, where *every paper* is spelled out and undergoes sideward movement from K to L. As is, the final structure in (44) cannot be linearized: given that the two instances of *every paper* are nondistinct, the preposition *after*, for instance, is subject to the contradictory requirement that it should precede and be preceded by *every paper*. In the cases discussed thus far, this kind of problem is remedied by trace deletion (deletion of lower chain links). However, trace deletion is inapplicable in (44): given that the two copies do not enter into a c-command relation, they cannot be identified as a chain. Thus, there is no convergent result arising from (44) and the parasitic gap construction in (45) is correctly ruled out.

(43) a. K = $[_{PP}$ after reading $[_{every}$ <every, paper> $]^1$]
b. L = $[_{VP}$ filed $[_{every}$ <every, paper> $]^1$]

(44) $[_{TP}$ John $[_{vP}\ [_{vP}$ filed $[_{every}$ <every, paper> $]^1$] $[_{after}$ <after, reading, $[_{every}$ <every, paper> $]^i$>]]]

(45) *John filed every paper without reading.

To sum up, the analysis explored above is very much in consonance with minimalist guidelines in that it attempts to derive construction-specific properties from general bare output conditions (more precisely, PF linearization), it limits the search space for deletion of copies (it can only happen within a c-command path), and it does not resort to the noninterface level of S-structure to rule out (45), like standard GB analyses do (see Chomsky 1982, for instance).[16] With respect to the main topic of this paper,

[16] It is not our intention here to present an analysis for all the different aspects involved in parasitic gap constructions. The aim of the discussion of the so-called S-structure licensing condition on parasitic gaps was simply to illustrate how sideward movement is constrained. See

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the lack of CED effects in acceptable parasitic gaps is argued to follow from the fact that Last Resort may license sideward movement from within a complex category XP, before XP is spelled out and its constituents become inaccessible to the Copy operation. In the next section, we will see that when parasitic gap constructions do exhibit CED effects, this is due to general properties of the system's design, which strives to reduce computational complexity.

4. Sideward Movement and Cyclic Access to the Numeration

Let us finally examine the unacceptable parasitic gaps constructions in (46), which illustrate the fact that parasitic gaps are not completely immune to CED effects.

(46) a. *Which book did you finally read after leaving the bookstore without finding?
b. *Which politician did you criticize before pictures of upset the voters?

Under one derivational route, the explanation for the unacceptability of the sentences in (46) is straightforward. The PP adjunct headed by *without* in (46a), for instance, must be spelled out before merging with the vP related to *leaving*, as represented in the simplified structure in (47a); hence, the constituents of this PP adjunct are not accessible to the computational system and sideward movement of *which book* from K to L is impossible. Likewise, sideward movement of *which politician* from X in (48a) to Y in (48b) cannot take place because the subject in (48a) has been spelled out and its constituent terms are inaccessible for copying; hence, the unacceptability of (46b).

(47) a. K = [leaving the bookstore [$_{without}$ <without, PRO, finding, which, book>]
b. L = read

(48) a. X = [$_{IP}$ [$_{pictures}$ < pictures, of, which, politician>] upset the voters]
b. Y = criticize

This account of the unacceptability of the parasitic gap constructions in (46) has crucially assumed that the computation proceeds from a "subordinated" to a "subordinating" derivational workspace; in all the cases discussed so far, sideward movement has proceeded from within an adjunct or subject to the object position of a subordinating verb. This assumption is by no means innocent. In principle, the computational system

Nunes 1995, 1998, Hornstein 1998, and Hornstein & Nunes 1999 for explanations for other properties of parasitic gap constructions under a sideward movement approach.

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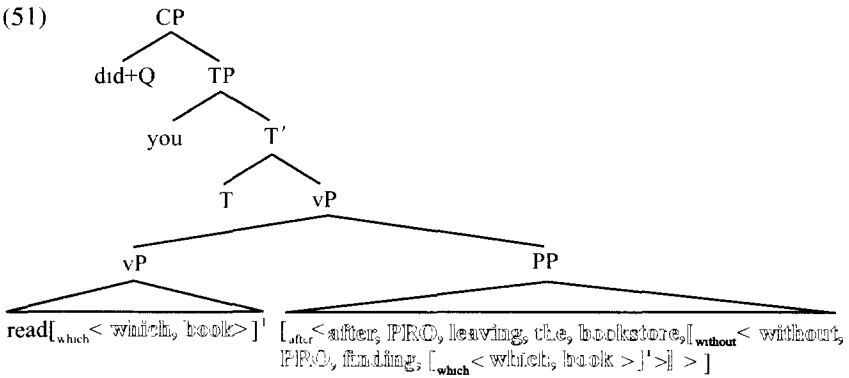
could also allow sideward movement to proceed from a "subordinating" to a "subordinated" derivational workspace, while still adhering to cyclicity. Suppose, for instance, that we assemble the matrix VP of (46a), before building the VP headed by *finding*, as represented in (49).

(49) a. K = [read [which book]]
b. L = finding

Given the stage in (49), *which book* could undergo sideward movement from K to L, and M in (50b) would be formed (irrelevant details omitted). Further computations after M was spelled out and merged with K would then yield the (simplified) structure in (51).

(50) a. K = [read $[_{which}$ <which, book> $]^1$]
b. M = [after PRO leaving the bookstore $[_{without}$ <without, PRO, finding, $[_{which}$ <which, book> $]^1$ >]]

(51)



The relevant aspect of (51) is that, although the *wh*-copy inside PP is not accessible to the computational system, the *wh*-copy in the object position of *read* is. It could then move to check the strong feature of Q and deletion of the lower *wh*-copies would yield the (simplified) structure in (52), which should surface as (46a).

(52) $[_{CP}$ $[_{which}$ <which, book> $]^1$ did+Q $[_{TP}$ you $[_{vP}$ $[_{vP}$ read $[$~~which~~ ~~<which, book>~~ $]^1$] $[_{after}$ <after, PRO, leaving, the, bookstore, $[_{without}$ <without, PRO, finding, $[$~~which~~ ~~<which, book>~~ $]^1$ >] >]]]]

Thus, if sideward movement were allowed to proceed along the lines of (49)–(50), where a given constituent moves from a derivational workspace W_1 to a derivational workspace W_2 that will end up being embedded under W_1, there should never be any CED effect in parasitic gap constructions and

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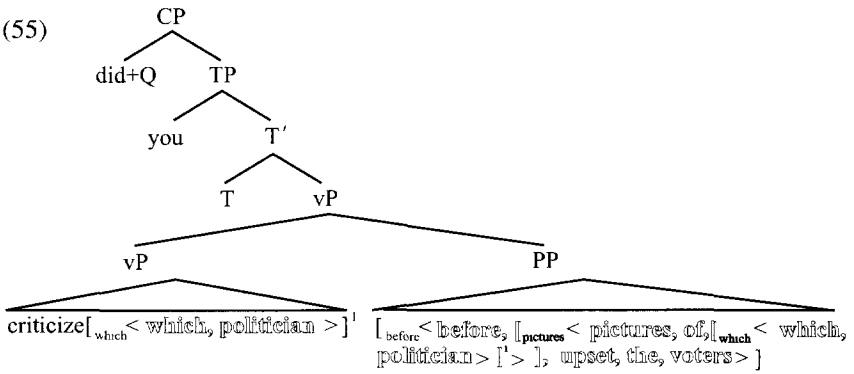
we would incorrectly predict that (46a) should be acceptable.

Similar considerations apply to the alternative derivation of (46b) sketched in (53)–(56). In (53)–(54), *which politician* moves from the object position of *criticize* to the complement position of the preposition. Further (cyclic) computations then yield the (simplified) structure in (55), in which the *wh*-copy in the matrix object position is still accessible to the computational system, thus being able to move and check the strong feature of Q. After this movement takes place, the whole structure is spelled out and the lower copies of *which politician* are deleted in the phonological component, as shown in (56). The derivation outlined in (53)–(56) therefore incorrectly rules in the unacceptable parasitic gap in (46b).

(53) a. X = [criticize [which politician]]
b. Y = of

(54) a. X = [criticize [$_{which}$ <which, politician>]^{1}]
b. Z = [of [$_{which}$ <which, politician>]^{1}]

(55)



(56) [$_{CP}$ [$_{which}$ <which, politician>]^{1} did+Q [$_{TP}$ you [$_{vP}$ [$_{vP}$ criticize ~~[$_{which}$ <which, politician>]^{1}~~] [$_{before}$ <before, [$_{pictures}$ <pictures, of, ~~[$_{which}$ <which, politician>]^{1}~~ >], upset, the, voters>]]]

The generalization that arises from this discussion is that sideward movement from a derivational workspace W_1 to a derivational workspace W_2 yields licit results just in case W_1 will be embedded in W_2 at some derivational step. In the undesirable derivations sketched in (49)–(52) and (53)–(56), sideward movement has proceeded from the "matrix derivational workspace" to a subordinated one. Obviously, the question is how this generalization can be derived from independent considerations.

Abstractly, the problem we face here is no different from the one posed by economy computations involving expletive insertion in pairs such as (57), originally noted by Alec Marantz and Juan Romero. The two sentences in (57) share the same initial numeration; thus, if the computational system had

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access to the whole numeration, economy should favor insertion of *there* at the point where the structure in (58) has been assembled, incorrectly ruling out the derivation of the acceptable sentence in (57b).

(57) a. The fact is that there is someone in the room.
b. There is the fact that someone is in the room.
(58) [is someone in the room]

Addressing this and other similar issues, Chomsky (1998) proposes that rather than working with the numeration as a whole, the computational system works with subarrays of the numeration, each containing one instance of either a complementizer or a light verb. Furthermore, according to Chomsky's (1998) proposal, when a new subarray SA_i is selected, the vP or CP previously assembled based on subarray SA_k becomes frozen in the sense that no more checking or thematic relations may take place within it. Returning to the possibilities in (57), at the point where (58) is assembled, competition between insertion of *there* and movement of *someone* arises only if the active subarray feeding the derivation has an occurrence of the expletive; if it does not, as is the case of (57b), movement is the only option and the expletive is inserted later on, when another subarray is selected.

This strongly derivational approach has the relevant components for a principled account of why sideward movement must proceed from embedded to embedding contexts. If the computational system had access to the whole numeration, the derivation of the parasitic gap constructions in (46), for instance, could proceed either along the lines of (47) and (48) or along the lines of (49)–(52) and (53)–(56), yielding an undesirable result because the latter incorrectly predict that the sentences in (46) are acceptable. However, if the computational system works with one subarray at a time and if syntactic objects already assembled become frozen when a new subarray is selected, the unwanted derivations outlined in (49)–(52) and (53)–(56) are correctly excluded. Let us consider the details.

Assuming that numerations should be structured in terms of subarrays, the derivation in (49)–(52) should start with the numeration in (59) below, which contains the subarrays A–F, each determined by a light verb or a complementizer.

(59) $N = \{\{_A\ Q_1, \textit{did}_1\},$
$\{_B\ \textit{you}_1, \textit{finally}_1, v_1, \textit{read}_1, \textit{which}_1, \textit{book}_1, \textit{after}_1\},$
$\{_C\ C_1, T_1\},$
$\{_D\ PRO_1, v_1, \textit{leaving}_1, \textit{the}_1, \textit{bookstore}_1, \textit{without}_1\},$
$\{_E\ C_1, T_1\},$
$\{_F\ PRO_1, v_1, \textit{finding}_1\}\}$

The derivational step in (49), repeated here in (60), which would permit the undesirable instances of sideward movement, is illicit because it accesses a

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new subarray before it has used up the lexical items of the active subarray. More specifically, the derivational stage in (60) improperly accesses subarrays B and F of (59).[17]

(60) a. K = [read [which book]]
b. L = finding

Similarly, the step in (53), repeated here in (62), illicitly activates subarrays B and D of (61), which is the structured numeration that underlies the derivation in (53)–(56).

(61) N = {{$_A$ Q_1, *did*$_1$},
{$_B$ *you*$_1$, v_1, *criticize*$_1$, *which*$_1$, *politician*$_1$, *before*$_1$}
{$_C$ C_1, T_1},
{$_D$ *pictures*$_1$, *of*$_1$, v_1, *upset*$_1$, *the*$_1$, *voters*$_1$}}

(62) a. X = [criticize [which politician]]
b. Y = of

The problem with the derivations outlined in (49)–(52) and (53)–(56), therefore, is not the instances of sideward movement themselves, but rather the derivational steps that should allow them. By contrast, lexical access in the derivational routes sketched in (47) and (48), repeated below in (64) and (66), may proceed in a cyclic fashion from the structured numerations in (63) and (65), respectively, without improperly activating more than one subarray at a time. However, as discussed above, sideward movement of *which book* in (64) or *which politician* in (66) is impossible because these elements have already been spelled out and are not accessible to the computational system.

(63) N = {{$_A$ Q_1, *did*$_1$},
{$_B$ *you*$_1$, *finally*$_1$, v_1, *read*$_1$, *after*$_1$},
{$_C$ C_1, T_1},
{$_D$ PRO_1, v_1, *leaving*$_1$, *the*$_1$, *bookstore*$_1$, *without*$_1$},
{$_F$ PRO_1, v_1, *finding*$_1$, *which*$_1$, *book*$_1$}}

(64) a. K = [$_{CP}$ C [$_{TP}$ PRO T [$_{vP}$ [$_{vP}$ leaving+v the bookstore] [$_{without}$ <without, C, PRO, T, finding+v, which, book>]]]]
b. L = read

[17] Following Chomsky 1998, we are assuming, largely for concreteness, that the maximal projection determined by a subarray is either vP or CP (a *phase* in Chomsky's 1998 terms). In convergent derivations, prepositions that select clausal complements must then belong to the "subordinating" array, and not to an array associated with the complement clause (otherwise, we would have a PP phase) Hence, the prepositions *after* and *without* in (59) and *before* in (61) belong to subarrays determined by a light verb, and not by a complementizer.

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(65) N = {{$_A$ Q_1, *did*$_1$},
{$_B$ *you*$_1$, v_1, *criticize*$_1$, *before*$_1$}
{$_C$ C_1, T_1},
{$_D$ *pictures*$_1$, *of*$_1$, *which*$_1$, *politician*$_1$, v_1, *upset*$_1$, *the*$_1$, *voters*$_1$}}

(66) a. X = [$_{CP}$ C [$_{TP}$ [$_{pictures}$ < pictures, of, which, politician>] T [$_{vP}$ [$_{pictures}$ < pictures, of, which, politician> [$_{v'}$ upset+v the voters]]]]
b. Y = criticize

The analysis of CED effects in parasitic gap constructions developed here can therefore be understood as providing evidence for a strongly derivational system, where even lexical access proceeds in a cyclic fashion.[18]

5. Conclusion

This paper has attempted to provide a minimalist analysis of classical extraction domains, in terms of derivational dynamics in a cyclic system. The main lines of research that provide a solution to the relevant kind of islands are (i) a computational system with multiple applications of Spell-out; and (ii) a decomposition of the Move operation into its constituent parts, taking seriously the idea that separate copies are real objects and can be manipulated in separate derivational workspaces (sideward movement).

Extraction domains are opaque because, after Spell-out, the constituent terms of a given chunk of structure, while interpretable, are no longer accessible to the rest of the derivation. At the same time, said opacity can be bypassed if an extra copy of the moving term manages to arise before the structure containing it is spelled out—something that the system in principle allows. However, this possibility is severely limited by other computational considerations. For example, Last Resort imposes that the extra copy be legitimated, which separates instances where this copy is made with no purpose other than escaping an island (a CED effect) from instances where the copy is made in order to satisfy a θ-relation (a parasitic gap construction). In the second case, the crucial copy can be legitimated prior to the Spell-out of the would-be island, thus resulting in a grammatical structure. Moreover, we have shown how sideward movement can only proceed, as it were, forward within the derivational history. That result is straightforwardly achieved in a radically derivational system, where the very access to the initial lexical array is done in a strictly cyclic fashion.

Although we find these results rather interesting, we do not want to finish without pointing out some of our worries, as topics for further research. Our whole analysis relies on the assumptions that copies are real and, as such, can be manipulated as bona fide terms within the derivation. If so, it is perplexing that, for the purposes of linearization different copies count as one, which

[18] For further evidence that sideward movement must proceed in this strongly derivational fashion, see Hornstein 1998 and Hornstein & Nunes 1999

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drives a good part of the logic of the paper. Of course, we can make this be the case by stipulating a definition of identity, as we have (token in the numeration as opposed to occurrence in the derivation); but we do not know why that definition holds. Second, it is fundamental for the account of island effects that spelled-out chunks be inaccessible to computation. However, chain identification can proceed across spelled-out portions, also in a rather surprising way. Once again, we can make things work by making c-command insensitive to anything other than the notion of containment; but we do not know why that should be or why c-command should hold, to start with, of chains. Finally, it should be noted that cyclic access to the numeration is key to keeping the proper order of operations; we have no idea why the relevant derivational cycles should be the ones we have assumed, following Chomsky (1998). All we can say with regard to all these questions is that we have suspended our disbelief, just to see how far the system can proceed within assumptions that are familiar.

References

BOBALJIK, J. & S. BROWN. 1997. Inter-arboreal operations: Head-movement and the extension requirement. *Linguistic Inquiry* 28:345–356.

BRODY, M. 1995. *Lexico-Logical Form: A radical minimalist theory*. Cambridge, Mass.: MIT Press.

CATTELL, R. 1976. Constraints on movement rules. *Language* 52:18–50.

CHOMSKY, N. 1982. *Some concepts and consequences of the theory of government and binding*. Cambridge, Mass.: MIT Press.

CHOMSKY, N. 1986. *Barriers*. Cambridge, Mass.: MIT Press.

CHOMSKY, N. 1995. *The Minimalist Program*. Cambridge, Mass.: MIT Press.

CHOMSKY, N. 1998. *Minimalist inquiries: The framework*. MIT Occasional Papers in Linguistics 15. Cambridge, Mass.: MITWPL.

CONTRERAS, H. 1984. A note on parasitic gaps. *Linguistic Inquiry* 15:704–713.

EPSTEIN, S. D. 1995. Un-principled syntax and the derivation of syntactic relations. Ms., Harvard University, Cambridge, Mass. [Pub. in *Working minimalism*, ed. S. D. Epstein and N. Hornstein, 317–345. Cambridge, Mass.: MIT Press, 1999.]

HIGGINBOTHAM, J. 1983. A note on phrase-markers. *Revue Quebecoise de Linguistique* 13:147–166.

HORNSTEIN, N. 1998. Move. Ms., University of Maryland at College Park.

HORNSTEIN, N. & J. NUNES. 1999. Asymmetries between parasitic gap and across-the-board extraction constructions. Ms., University of Maryland at College Park and Universidade Estadual de Campinas.

HUANG, C.-T. J. 1982. Logical relations in Chinese and the theory of grammar. Ph.D. dissertation, MIT, Cambridge, Mass.

KAYNE, R. 1984. *Connectedness and binary branching*. Dordrecht: Foris.

KAYNE, R. 1994. *The antisymmetry of syntax*. Cambridge, Mass.: MIT Press.

KITAHARA, H. 1997. *Elementary operations and optimal derivations*. Cambridge, Mass.: MIT Press.

LARSON, R. 1988. On the double-object construction. *Linguistic Inquiry* 19:335–391.

MUNN, A. 1994. A minimalist account of reconstruction asymmetries. In *Proceedings of the North East Linguistic Society 24*, ed. M. Gonzàlez, 397–410. Amherst, Mass.: GLSA.

NUNES, J. 1995. The copy theory of movement and linearization of chains in the

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minimalist program. Ph.D. dissertation, University of Maryland at College Park.
NUNES, J. 1998. Sideward movement and linearization of chains in the minimalist program. Ms., Universidade Estadual de Campinas.
NUNES, J. 1999. Linearization of chains and phonetic realization of chain links. In *Working minimalism*, ed. S. Epstein and N. Hornstein, 217–250. Cambridge, Mass.: MIT Press.
URIAGEREKA, J. 1998. *Rhyme and reason: An introduction to minimalist syntax.* Cambridge, Mass.: MIT Press.
URIAGEREKA, J. 1999. Multiple Spell-out. In *Working minimalism*, ed. S. Epstein and N. Hornstein, 251–282. Cambridge, Mass.: MIT Press.

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